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**Distribution of the habitat suitability of the main malaria vector in French Guiana using
Maximum Entropy modeling**

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Abstract

Malaria is an important health issue in French Guiana. Its principal mosquito vector in this region is *Anopheles darlingi*. Knowledge of the spatial distribution of this species is still very incomplete due to the extent of French Guiana and the difficulty to access most of the territory.

Species Distribution Modeling based on the maximal entropy procedure was used to predict the spatial distribution of *An. darlingi* using 39 presence sites.

The resulting model provided significantly high prediction performances (mean 10-fold cross-validated partial AUC and continuous Boyce index equal to, respectively, 1.11 – with a level of omission error of 20 % – and 0.42). The model also provided a habitat suitability map and environmental response curves in accordance with the known entomological situation.

Several environmental characteristics that had a positive correlation with the presence of *An. darlingi* were highlighted: non-permanent anthropogenic changes of the natural environment; the presence of roads and tracks; opening of the forest. Some geomorphological landforms and high altitude landscapes appear to be unsuitable for *An. darlingi*.

The Species Distribution Modeling was able to reliably predict the distribution of suitable habitats for *An. darlingi* in French Guiana. Results allowed completion of the knowledge of the spatial distribution of the principal malaria vector in this Amazonian region, and identification of the main factors that favor its presence. They should contribute to the definition of a necessary targeted vector control strategy in a malaria pre-elimination stage, and allow extrapolation of the acquired knowledge to other Amazonian or malaria-endemic contexts.

Keywords: Maxent, species distribution model, presence-only, *Anopheles darlingi*, sampling bias

Malaria is a public health issue in the Amazonian region, with major transmission foci depending on specific local characteristics associated with changing environmental and socio-demographic contexts. French Guiana is a French overseas territory with ~260,000 inhabitants. It remains one of the major malaria foci in the region, despite an improving epidemiological situation during the past ten years. The number of reported clinical cases has significantly dropped from 4,479 in 2005 to 434 in 2015 (Petit-Sinturel et al. 2016), and now corresponds to an incidence rate of two cases for 1,000 inhabitants for the whole territory, making it possible to target the pre-elimination of the disease in 2018 (Agence Régionale de Santé Guyane 2015). *Plasmodium vivax* is at present predominant and this species was responsible for 67% of the diagnosed cases of malaria in the territory in 2014, the others being mainly due to *Plasmodium falciparum* (Musset et al. 2014, Ardillon et al. 2015). However, this epidemiological situation is heterogeneous in space and time. In particular, a recrudescence of malaria cases is currently observed in the inland region (Saül, Cacao, and Régina) and eastern French Guiana (municipalities of Camopi and Saint-Georges-de-l'Oyapock), with a general incidence rate reaching 55.2 cases per 1,000 inhabitants in 2013 (Musset et al. 2014), likely due to the emergence and/or persistence of local foci of high malaria transmission (Berger et al. 2012, Musset et al. 2014). This Amazonian region, especially near the international borders, includes vulnerable populations. Some are hard-to-reach and have poor access to health services and treatment-seeking behaviors that may favor the development of resistance to antimalarial drugs (Musset et al. 2014, Wangdi et al. 2015). Uncontrolled areas of malaria transmission are also prevalent in illegal gold mining areas (Pommier de Santi et al. 2016a, Pommier de Santi et al. 2016b). The epidemiological situation remains quite unstable, and pre-elimination of malaria, corresponding to an incidence rate below one case for 1,000 inhabitants in any locality of French Guiana, remains a major challenge.

In this context, public health authorities must maintain control efforts while targeting them more precisely and objectively in time and space (Alimi et al. 2015). A map of malaria risk in French

Guiana is updated regularly by the regional unit of the French National Public Health Agency, based on the number of cases reported per locality and the data available on movements of human populations at risk, especially due to gold mining activities. This map is validated by the local expert committee of epidemic diseases (Comité d'Experts des Maladies à Caractère Épidémique, CEMCE), which brings together different experts of the disease in the region (from the Health Surveillance Agency, the Pasteur Institute of French Guiana, the Regional Unit of the National Public Health Agency, vector control services, hospitals, and other diagnosis and care centers, and the Defense Health Service in French Guiana). The lack of objective knowledge of several key factors, especially the spatiotemporal distribution of the main malaria vectors and human populations infected by *Plasmodium* and/or carrying gametocytes, makes such a map highly approximate.

Anopheles (Nyssorhynchus) darlingi Root (Diptera: Culicidae) is one of the most efficient malaria vectors in South America and is considered to be the primary malaria vector in French Guiana because of its anthropophilic behavior, natural infectability, high density, and sensitivity to *P. falciparum* (Girod et al. 2008, Hiwat et al. 2010, Fouque et al. 2010). Used entomological data collection for the entire territory, for the mapping of entomological risk indicators at the regional scale, is not feasible. French Guiana occupies a large territory (84,000 km²) which is mostly covered by rain forest (more than 80%) and highly inaccessible. Knowledge of the recent geographical distribution of *An. darlingi* is thus restricted to coastal areas, some villages along the international border rivers, and some illegal gold mining sites (Figure 1).

Species Distribution Modeling (SDM) offers an efficient solution to geographically extrapolate such knowledge to the entire territory (Pearson et al. 2007). Species Distribution Modeling produces maps of species habitat suitability by using known presence locations of the species and relevant environmental data. The use of SDM is thus encouraged to “improve and facilitate the development of alternative vector control strategies” (Alimi et al. 2015). Numerous SDM approaches are

proposed in the literature. Some of them, such as Maximum Entropy (Maxent; Phillips et al. 2006), Genetic Algorithm for Rule-Set Prediction (GARP; Stockwell 1999), Boosted Regression Trees (BRT; Friedman et al. 2000), Generalized linear and additive models (GLM and GAM; Guisan, et al. 2002), and Multivariate adaptive regression splines (MARS; Leathwick et al. 2005), exploit only species presence information, offering a significant advantage over methods that also require absence data. Indeed, absence data are often difficult to obtain. According to Peterson (2007) and Hirzel et al. (2002), absence can result from (1) the non-detection of the species in a suitable habitat, even if it is present, (2) the actual absence of the species for historical reasons, whereas the habitat is suitable, and (3) the true absence of the species and the unsuitability of the habitat. Comparative studies (Elith et al. 2006, Tognelli et al. 2009, Pearson et al. 2007, Hernandez et al. 2006, Wisz et al. 2008) show that Maxent is able to fit complex functions between habitat suitability and predictor variables, is the least sensitive to the size of the presence dataset, and tends to outperform other comparable methods when the dataset is small. In this study, the mapping of the habitat suitability of *An. darlingi* at the scale of all of French Guiana was performed using the Maxent SDM approach. This work aims to provide reliable maps for improving malaria transmission risk mapping in French Guiana, and to identify the environmental factors and associated mechanisms that favor the presence of *An. darlingi*.

Materials and Methods

Study area

French Guiana (84,000 km²), a French overseas region located in South America, is separated from Suriname by the Maroni River and from Brazil by the Oyapock River and the Tumuc-Humac mountains. More than 80% of the territory is covered by rain forest. The country has an equatorial climate characterized by two annual dry seasons, from mid-August to mid-November and in March, and two wet seasons, from mid-April to mid-August and mid-November to February. The average annual rainfall reaches 4,000 mm and 2,000 mm in the wettest (north-east) and driest (north-west)

areas, respectively (Hammond 2005). The average monthly rainfall is >100 mm for the entire territory throughout the year, except for the three driest months: September, October, and November (Héritier, 2011). The average humidity is between 80% and 90%. The temperature is homogeneous over the entire territory throughout the year, with an average annual temperature of 26°C. The difference between the minimum and maximum daily temperature is more important than the annual variations. For example, in Maripasoula (on the border with Surinam) and Camopi (on the border with Brazil), the annual ranges of the minimum and maximum temperatures were, 4.3°C and 9.6°C (averages over the period 2001-2008), respectively, whereas the mean daily thermal amplitude was 9.8°C (average over the period 2001-2008; Météo-France, 2016). The population of ~260,000 inhabitants is unequally distributed throughout the territory. Approximately 90% of the population lives in the coastal area and most of the rest lives along the Maroni and the Oyapock rivers (Amerindians and Bush-Negroes). However many people live and/or transit through inland and remote areas of the territory (forestry workers, gold miners, and soldiers). According to many studies (Berger et al. 2012, Verret et al. 2006, Queyriaux et al. 2011, Hustache et al. 2007, Stefani et al. 2011, Pommier de Santi et al. 2016a), Amerindians, gold miners, and soldiers may be highly infected by malaria, whereas the areas in which they live and/or transit are those with the poorest knowledge of the presence and density of malaria vectors. It is thus of potential interest to consider the malaria risk, study the distribution of malaria vectors, and implement prevention and control actions over the entire territory of French Guiana.

Species Records

Presence sites of *An. darlingi* were provided by surveys of the Medical Entomology Unit of the Pasteur Institute of French Guiana and the Defense Health Service in French Guiana. *Culicidae* collections were performed using either human landing catches or traps (light traps or odor baited traps). Human landing catches consisted of exposing collector's lower leg and collecting landing mosquito with a mouth aspirator. Collectors were members of the Pasteur Institute or Defense

Health Services, they were aware of the risks associated with the method and had given their free consent. Malaria prophylaxis was proposed and information on the medication was explained. Light trap catches were performed with Center for Disease Control and Prevention (CDC) light traps, and odor baited catches were performed with Mosquito Magnet ® traps (Woodstream Corporation, Lititz, PA) baited with Octenol, a combination considered to be the best candidate for *Anopheles* surveillance in the region (Vezenegho et al., 2015).

Anopheles species were morphologically identified using taxonomic keys specific for the region (Floch and Abonnenc 1951, Faran and Linthicum 1981, Forattini 1962). Only *Culicidae* collections performed since the year 2000 were precisely geolocated by GPS coordinates and were used for the study (Figure 1). These data correspond to 74 capture sites for the family *Culicidae*, and to 48 presence sites for the species *An. darlingi*.

The difficulty in accessing most of the French Guiana territory, and the priority given to the areas at risk of malaria transmission where many people live, led to a significant sampling bias with oversampling of the anthropized region of the territory, notably those easily accessible by roads (Figure 1).

Ecological knowledge and hypotheses

The presence of *An. darlingi* is linked to compositional and configurational features of the land cover and land use, as they partially determine breeding, feeding, and resting sites of the vector (Stefani et al. 2013). The natural environment for this vector in the Amazonian region includes floodable savanna, swamps (Girod et al. 2011, Zeilhofer et al. 2007), and flooded forest (Rozendaal 1992). Larvae are found along river edges, on flooded riverbanks, creeks, and pools formed near river-beds (Rozendaal 1992, Hiwat et al. 2010). Breeding sites are generally situated at low altitude (Mouchet 2004) and solely in freshwater, as *An. darlingi* is sensitive to salinity (Deane et al. 1948). Hydrological and geomorphological factors are responsible for the formation and destruction of *Anopheles* breeding sites (Smith et al. 2013).

Human activities, comprising deforestation and fish farming, also contribute to the creation of active breeding sites (Patz et al. 2000, Richard 1987, Stefani et al. 2013, Takken et al. 2005, Terrazas et al. 2015, Vittor et al. 2006, Vittor et al. 2009). Unpaved roads, tracks, and culverts form ideal breeding sites for *An. darlingi* in the Amazon region (Singer and Castro 2001). The presence of *An. darlingi* is also maintained by regular human presence due to its strong anthropophilic behavior. However, the presence and density of *An. darlingi* can either be favored or restricted depending on the type and intensity of the anthropogenic impacts. Stefani et al. (2013) systematically reviewed the literature and showed that all the studies describe the same mechanisms linking deforestation, land use, and the degree of urbanization with malaria transmission risk in the Amazonian region: opening the forest and maintaining a high degree of interaction between forested and deforested areas decreases the distance between feeding, breeding, and resting sites of *An. darlingi*, favoring the presence and high density of the vector (as well as a high probability of contact between humans and vectors); in contrast, intensifying deforestation and creating large urbanized and/or cultivated surfaces tends to decrease suitable habitat for *An. darlingi*. These two antagonistic consequences of human activities were considered in the SDM described here, by explicitly separating favorable and unfavorable factors in the environmental characterization. The optimum temperature range for *An. darlingi* is between 20 and 30°C with a humidity of above 60% (Martens et al. 1995). Several studies established a minimal monthly rainfall threshold to designate suitable breeding habitats for *Anopheles* (reviewed in Smith et al. 2013). These values vary between 10 and 80 mm and need to be maintained for three or four months.

Environmental Variables

Environmental variables chosen as SDM inputs must characterize the ecological factors that influence the presence of *An. darlingi*, previously described. These factors are separated into three types: 1) natural environment features, associated with land cover, land use, and geomorphology for which the impact on the presence of *An. darlingi* depends on specific values or categorical classes;

2) anthropogenic activities that non-permanently alter the natural environment on a highly local scale and favor the presence of *An. darlingi*; 3) urbanization, corresponding to human presence and activities that permanently alter the natural environment over large areas and hinders the presence of the vector. Meteorological variables were not included in the model, because the temperature, rainfall, and humidity fall within the optimal ranges for presence of the species in French Guiana. Thus, these variables cannot significantly explain differences in the time average habitat suitability distribution over the year (this point is extensively discussed in the Discussion section).

Raw Geographic Data. Variables chosen as SDM inputs were derived from the following raw geographic data:

- Geomorphological landscape (*GLS*) and Geomorphological landforms (*GLF*) from the French Forest Office (ONF) (Guitet et al. 2013);
- Landscape types (*LS*) from the French Agricultural Research Centre for International Development (CIRAD) (Gond et al. 2011). This provides the distribution of landscape types in French Guiana, most being forested landscapes;
- Altitude (*ATL*) derived from the Digital Elevation Model provided by the Shuttle Radar Topography Mission (SRTM, spatial resolution: 30 meters) of the United States Aeronautics and Space Administration (NASA);
- Human footprint (*HFP*): An integrated human activity index that gives a general measure of the extent of expected threats on biodiversity, by assigning a score depending on the nature of the disturbance. It combines sublayers spatializing human population density, urban areas, legal and illegal mining sites, agriculture, forest settlements and camps, tourist camps, logged areas (forest activities), and potential hunting areas corresponding to a zone of two kilometers around roads, tracks and rivers, likely to be used by humans. The total disturbance score is the sum of all human activity scores (de Thoisy et al. 2010);
- Roads and tracks from the BD TOPO® database of the French Institute of Geographical

and Forestry Information (IGN).

Table 1 summarizes the main features of these raw geographic data.

Definition of Environmental Variables Used as Inputs for SDM. Several variables were

extracted from the previously described raw data to better reflect the ecological knowledge and hypotheses mentioned above. The reference spatial resolution (pixel size) permitting the integration of all environmental layers was set to 1 by 1 km, *i.e.*, the coarsest resolution of the available layers, associated with the *LS* map.

The length of roads and tracks outside of urban areas (*ROADS*) was computed in the 1 km-cell grid from the BD TOPO® database.

The sublayers composing the *HFP* were first rasterized into 30-m grid cells, the smaller polygon of the *HFP* having a size of approximately 40 by 10 m. Distinct attributes were then extracted:

- The percentage of urbanization (*PER_URB*) within the 1 km grid cells;
- The percentage of urbanization within the eight neighbor cells of each urban cell (*PER_URB_NEIGH*), which permits distinguishing small from large urban areas. This layer was obtained for each 1 km-cell considered to be urban (*i.e.*, with $PER_URB \geq 50\%$), by averaging the *PER_URB* values for the eight (1 km side) neighbor pixels;
- The human activities which non-permanently alter the natural environment (*HA*), by first summing the scores of the following sublayers: tourist and forest camps, mining activities and logged areas, hunting areas nearby rivers, and then, by computing the minimum, median, and maximum values within the 1-km grid cells.

The agriculture sublayer from *HFP* was not used because it covers only the coastal area. The population density sublayer was also excluded because it did not have sufficient level of detail. The sublayer of potential hunting areas near roads and tracks were not used to avoid duplication of the length of roads and tracks outside of urban areas computed previously.

For each 1-km grid cell, the majority class of the categorical variables *GLS* and *GLF*, and the

minimum, median, and maximum altitude (*ALT*) values were computed. Eventually, some corrections of the *LS* layer were performed as it did not identify urban areas and did not distinguish flooded forests associated with freshwater from those of the coastal strip associated with brackish water (mangroves): *LS* cells with an *PER_URB* value greater than or equal to 50% were reclassified into a new *LS* class referred to as *Urban*; *LS* cells classified as *Flooded forest* and corresponding to mangroves according to the coastal land use map provided by the ONF (Office National des Forêts Direction Régionale de Guyane, 2013) were reclassified as *Mangrove*. The variable *PER_URB* was excluded from the input SDM variables, as the urban areas were mapped, and their extent quantified, by the corrected *LS* and *PER_URB_NEIGH* layers, respectively. Table 1 lists and describes the environmental variables used to build the model.

Maxent Model Principle

Maxent is an SDM which requires environmental variables and species presence-only data. It is based on the principle of maximum entropy to estimate an (*a priori*) unknown probability distribution over the entire study area. This probability distribution assigns a value that is proportional to the probability of the presence of the species to each pixel of the study area. It is therefore interpreted as a habitat suitability index (HSI) across the study area (Phillips et al. 2006). The Maximum Entropy principle consists of approximating the unknown probability distribution by finding the one that maximizes entropy and satisfies the constraints imposed by the environmental features at the known sites of presence. Environmental features are a set of input environmental variables chosen according to their expected relevance for the studied taxon (Phillips et al. 2006, Elith et al. 2011). The constraints ensure that the environmental values expected under the approximated probability distribution are consistent with environmental information observed at the presence points.

In practice, the Maxent distribution is defined on a set of points called background points. These

points should reflect the available environmental conditions of the study area and are chosen by uniform random sampling. This approach assumes that the presence data are not biased and that environmental conditions are uniformly sampled (Yackulic et al. 2013). However, in practice, some areas are more intensively sampled than others, and environmental conditions are not uniformly distributed and may imply a strong sampling bias. Phillips et al. (2009) proposed selecting the background points with the same environmental bias as the presence dataset to correct the effect of this sampling bias.

Model Building and Evaluation

Eleven environmental variables and 48 *An. darlingi* presence points (their coordinates were in the table in supplementary material S1) are used as inputs for Maxent. Only one presence site was selected to build the model when more than one occurred in the same pixel. As a result, only 39 presence sites were actually used for building the model. Hinge and categorical features were selected for the environmental variables. A hinge feature provides a good compromise between simplicity and the quality of the approximation of the species response curves (Elith et al. 2011, Phillips and Dudík 2008).

In this study, the distribution of the background points was biased so that the selection bias corresponds to that of the sampling. The sampling bias was defined as the relative sampling effort in the environmental space, and was estimated by considering the capture locations of *Culicidae*, obtained using the same capture techniques and supposed to be subjected to the same sampling bias as the *An. darlingi* species. The details of the method to create the relative sampling effort map are described in supplementary material S2.

The model was computed using version 3.3.3k of Maxent. The recommended values derived from Phillips and Dudík (2008) concerning the regularization parameters and the background set size, were applied. Regularization parameter values were set to 0.25 and 0.5 for categorical and hinge features, respectively, and the size of the background was set to 10,000. The extrapolation option

was not selected to avoid making predictions in environmental domains in which the model was not trained. The model was fitted using the full data set and evaluated using a 10-fold cross-validation procedure. The Receiver Operating Characteristic (ROC) curves and the associated Areas Under the ROC Curve (AUC) were computed. This was completed by computing the mean partial AUC ratios (Peterson et al., 2008), consisting of the ratios of the partial AUCs of the model over the null AUC (corresponding to random prediction), for omission errors (E) of 20, 10, and 5%. The Continuous Boyce Index (CBI), considered to better adapted to presence-only models than the AUC (Hirzel et al., 2006), was also computed. The gain (regularized training gain) was also used to evaluate the performance of the model prediction. It is a measure of the likelihood of the sample, and indicates how much better the estimated distribution fits the presence points than the uniform distribution, which corresponds to a null gain (Yost et al. 2008).

The importance of each variable was estimated using two methods, a heuristic method and the jackknife test. The heuristic method computes the percentage contribution of each variable to the model. During the training process, the increase of the gain is due to the adjustment of the feature weights and this increase is assigned to the environmental variable that the feature depends on. The sum of these increases in gain indicates the percentage contribution of each environmental variable. The jackknife test evaluates the individual contribution of each variable to the model by estimating the difference of the gain when removing each variable, one by one, and when considering the given variable alone to build the model.

Results

The mean AUC was 0.93, and the mean partial AUC ratios were 1.08, 1.03, and 1.01 for maximum omission errors sets to 20, 10, and 5% respectively. The mean CBI was 0.356 and the mean gain was 3.14. Three variables cumulatively contributed >80% (Table 2): the length of roads and tracks outside of urban areas (*ROADS*), the percentage of urbanization of neighboring pixels (*PER_URB_NEIGH*), and landscape (*LS*). The maximum value of the human activities which non-

permanently alter the natural environment (*HA_MAX*), geomorphological landscape (*GLS*), minimum altitude (*ALT_MIN*), and geomorphological landform (*GLF*) contributed moderately to the model, with contributions of 6.84%, 5.35%, 1.34%, and 1.19%, respectively. The following input variables contributed very little to the model: minimum and median values of human activities which non-permanently alter the natural environment (*HA_MIN* and *HA_MED*; 0.35 and 0.24%, respectively); and median and maximum values of altitude (*ALT_MED* and *ALT_MAX*; 0.69 and 0.06%, respectively).

The results of the Jackknife test confirmed the non-significant contribution of the input variables *HA_MIN*, *HA_MED*, *ALT_MED*, and *ALT_MAX* (Table 2).

A second model was built using only the most highly contributing environmental variables: *ROADS*, *LS*, *PER_URB_NEIGH*, *HA_MAX*, *GLS*, *GLF*, and *ALT_MIN*. The overall performance of this simpler model was very similar to the previous one, with the mean AUC and partial AUC ratios equal to 0.93 and 1.11, 1.05, and 1.03, respectively. The mean gain was equal to 3.19 and the mean CBI was 0.421. Relative contributions of the input variables were also very similar (Table 3).

The response curves of the environmental variables are represented in Figures 2 and 3. They show that the HSI is maximal when the *PER_URB_NEIGH* is below 8%. Above this value, the HSI decreases progressively towards 0. The HSI increases as *ROADS* increases up to 7,000 meters, reaches a plateau value, and then tends to decrease above 10,000 meters. Among all *LS* classes, *Woodland savanna/dry forest* and *Open forest* contribute the most to the high HSI values. The geomorphological landscape classes *Coastal flat plain* and *Plain with residual relief* and the geomorphological landform classes *Small-size and flat wet land*, *Small-size rounded hill*, and *Lowered half-orange relief* – a tropical relief type corresponding to a hill with convex flanks giving to it a roughly hemispherical shape (George, 1972) and usually linked to flat or swampy lowlands drained by streams with meanders – are also associated with high HSI values. The HSI is maximal when *ALT* is ~0, with a rapid decrease as altitudes increase. The *HA_MAX* response curve presents a

more complicated profile. The HSI increases for *HA_MAX* values between 0 and 8, decreases until *HA_MAX* reaches 24, and then again increases as values continue to climb above 24.

The map of habitat suitability for *An. darlingi*, based on all the presence data for modeling, shows six main areas (A – F) with a high HSI and a seventh area (G) corresponding to an epidemiological interest area (see Figure 4). A qualitative analysis was performed to determine the characteristics of the environmental variables of the areas with high HSI values (Table 4).

In the coastal area (A), where 90% of the Guyanese population lives, the HSI tends to be higher along the main road representing the main traffic route in French Guiana. Focusing on the main urban areas, represented in Figure 5, the HSI values within the highly urbanized districts of Cayenne and Kourou (rectangles in Figure 5) are lower than those of the surrounding pixels that are not considered to be highly urbanized. A very high HSI was predicted within the urban area of Saint-Laurent-du-Maroni. However, none of the pixels characterizing this city has a *PER_URB_NEIGH* value higher than or equal to 50%. The high HSI values in areas B, D, E, and G are characterized by the environmental variables *ROADS*, *HA_MAX*, the classes *Open forest* and *Mixed high and open forest*, and flat or moderately hilly terrain. The high HSI in areas C and F is essentially linked to *Open forest* and flat terrain.

The areas for which the model did not predict the HSI, due to the choice to not extrapolate to environmental domains not used to train the model, correspond to areas with an altitude higher than 400 meters. They represent a small number of pixels of the study area.

Discussion

The prediction performances of the model are excellent and significantly greater than those of the null model. The following discussion focuses on the ecological interpretation of the results and the methodological choices and alternatives.

Environmental Factors Explaining the Habitat Suitability

The geographic distribution of habitat suitability is consistent with existing knowledge of the

entomological situation despite the small number of presence points. The high HSI values can be explained by different environmental contexts depending on the geographical locations. In most areas (A, B, D, and E), the HSI values depend on human presence and activities, characterized by the environmental variables *HA_MAX* and *ROADS* (in areas D, E, and B, most roads are not paved and correspond mostly to tracks). The significantly positive correlation between the variable *ROADS* and the HSI confirms that road and track opening, accompanied by deforestation and pooling of rainwater at the roadside, may favor breeding sites (Singer and Castro 2001). The response curve for the variable *ROADS* (Figure 3) reaches a plateau above 7,000 meters of road per square kilometer and decreases thereafter. The decrease of the HSI at values above 7,000 meters suggests that the density of the road network leads to an improvement of the road quality (paved road eliminating culverts, adding sidewalks), thus limiting the availability of breeding and/or resting sites, in the same way as urbanization. Indeed, the response curve of the *PER_URB_NEIGH* variable confirms that highly urbanized areas provide a poorly suitable habitat for *An. darlingi* (Figure 3). Intensive urbanization implies concrete paving, the decrease or removal of green areas and forests, and consequently, the destruction of breeding and resting sites for *An. darlingi* (Stefani et al. 2013). This phenomenon is observed in the highly urbanized areas of Cayenne and Kourou (Figure 5). In contrast, Saint-Laurent-du-Maroni, the second largest urban area of French Guiana in terms of urbanization size and density, has high HSI values. In fact, unlike Cayenne and Kourou, this area is not considered to be highly urbanized using the criterion of this study ($PER_URB_NEIGH \geq 50\%$). However, the result for Saint-Laurent-du-Maroni seems unlikely because the presence of *An. darlingi* has not yet been reported in an urban area. Further field works could confirm the presence of this species in the city. The sensitivity of the model for the criterion that defines a highly urbanized area may also merit further study.

The values of *HA_MAX* in areas D and E were essentially associated with mining activity. In French Guiana, this activity is responsible for forest loss reaching 2,000 hectares per year (Office National

des Forêts Direction Régionale de Guyane, 2014). Between 2001 and 2013, Alvarez-Berrios and Aide (2015) estimated that the largest forest loss due to gold mining in the tropical and subtropical moist forest in South America was situated in the Guianan region including French Guiana. This suggests that this activity, resulting in deforestation and creating sources of standing water such as mining pits, combined with the presence of a large number of people, creates suitable conditions for *An. darlingi*. The high HSI in these two areas is also explained by the *Mixed high and open forest* landscape which is associated with human disturbance (Gond et al. 2011). Indeed, this landscape is described as a forest environment linked to young or unstable vegetation mostly due to first stages of anthropization. These results confirm the important role of human presence in the creation of suitable habitats for *An. darlingi*, which is also consistent with the strong anthropophilic behavior of this vector.

Some landscape types which are not directly associated with human presence or activities were also associated with a HSI. The *Woodland savanna/dry forest* class appears to highly contribute to high HSI values (Figure 2). It corresponds to the driest landscape in French Guiana (Gond et al. 2011), but can be seasonally inundated due to its poor drainage, creating breeding sites (Rosa-Freitas et al. 2007). The high HSI values in this area are in accordance with previous studies (Vezenegho et al. 2015, Dusfour et al. 2013), which reported finding *An. darlingi* in the coastal savanna environments of French Guiana. In uninhabited areas (zones F and C in Figure 4), a high HSI is associated with the *Open forest* class (LS layer) and flat terrain. This LS class can be associated with different land cover types in French Guiana (Gond et al. 2011) depending on the geographical location. Consequently, this *LS* class may differentially affect *An. darlingi* habitat suitability. The *Open forest* in area C mainly corresponds to wetlands (classified as *Flooded forest* according to the coastal land use map provided by the Office National des Forêts Direction Régionale de Guyane, 2013), whereas in area F, it corresponds to *Large surfaces of bamboo thicket and forbs*. *Anopheles darlingi* was found in flooded forest; however, to our knowledge, no information is available concerning its

presence in large areas of bamboo thicket and forbs. The prediction in these areas should be taken with precaution as a more precise description of the habitats within *Open forest* class is required. Overall, this information highlights that natural environment could form highly suitable habitats despite the high anthropophily of *An. darlingi*.

Meteorological Variables

In this study, meteorological variables were not used to build the model. Temperatures fall within the optimal range for the species presence, and were considered to be geographically and temporally too homogeneous to explain differences in the spatial distribution of habitat suitability. Such a hypothesis is common in the Amazonian context. Olson et al. (2009) report that in their study region (Amazon basin), “monthly temperatures were between 24.6°C and 29.4°C (well within the range for optimal malaria transmission) for 95% of the observations,” and consequently did not include temperatures in their model. In French Guiana, several studies also used rainfall data to study the intra-annual variations in *An. darlingi* density (Hiwat et al., 2010, Girod et al., 2011). The exclusion of rainfall data is more debatable, as rainfall clearly influenced the intra-annual density of *An. darlingi* in the study region (Hiwat et al., 2010, Girod et al., 2011, Vezenegho et al., 2015) even if the relationship was not systematically observed (Girod et al., 2011). The evidence for this impact on densities is that *An. darlingi* habitat suitability varies at an intra-annual scale, due to the alternation of dry and wet seasons. However, the entire study area is subject to this alternation. Moreover, given the high density of the French Guiana hydrological network and that the driest area (north-west) still receives 2,000 mm a year, it can be reasonably assumed that *An. darlingi* can find suitable conditions within the entire territory throughout most of the year. In French Guiana, the geomorphological landscape highly influences the availability of breeding sites, and therefore their spatial distribution, whereas the rainfall quantities influence the intra-annual variations of *An. darlingi* densities. As a consequence, on an average over the year, we assume that the significant factor influencing the distribution of habitat suitability is not the quantity of rainfall, but the

capacity of the landscape to provide suitable breeding sites when it rains.

Model Parametrization

The model was run by using the regularization parameter values and background set size recommended by Phillips and Dudík (2008), instead of those determined from specific experiments, as suggested by Merow et al. (2013). Phillips and Dudík (2008) tested a set of regularization parameter values with 48 species datasets that contained 11 to 13 environmental variables and a small number of categorical variables (1-3, as they considered discrete ordinal variables to be categorical). Nine of these datasets contained between 30 and 60 occurrences. The characteristics of the dataset exploited in our study (39 occurrence records; 13 and seven environmental variables including three categorical ones) are assumed to be quite similar of those of the datasets used by Phillips and Dudík (2008). We thus assumed that the pseudo-optimal parameters proposed by Phillips and Dudík (2008) could be confidently used in our study. Similarly, the background size was set to 10,000 based on the tests realized by Phillips and Dudík (2008), with 226 species and a median number of 57 presence sites. Better prediction performance may have been obtained by tuning the regularization values and background size and adding input environmental variables and features. However, the risk would have been to favor overfitting to the detriment of the bio-ecological interpretation of the model (see for example Merow et al., 2013). According to the entomologists who participated in the study, the model appears to be a good compromise between overfitting (that would have predicted suitable areas near occurrence points only) and being too general (that would have predicted suitable areas in too many environmental contexts for which the specialists have no species presence evidence).

Correction of the sampling bias effect

In this study, the effect of sampling bias was corrected by selecting background points with the same environmental bias as the sampled points. This approach appeared to be useful when applied to *An. darlingi* in French Guiana. Without a bias effect correction, the model predicted very high

HSI values in highly urbanized areas whereas these areas are known to be unsuitable for this vector (see above). The biased background set is more concentrated around the sampled points (in the environmental space) than the uniform random background, and is not likely to include environmental conditions that are highly dissimilar to those encountered at the sampled points. As a result, environmental conditions highly dissimilar to those of the sampled points can be subjected to extrapolation, which may lead to erroneous habitat suitability predictions and bio-ecological interpretations. This justifies not using the extrapolation option for modeling. The predicted HSI map from the model with a biased background contains several excluded areas, whereas that of the model with a uniform random background does not. Excluded areas correspond to high altitude areas which are unsuitable for *An. darlingi* (Mouchet 2004).

When using a uniform random background, the three most contributive variables (cumulative contribution equal to 85.5%) were all directly linked to human presence and territory accessibility (*ROADS*: 38.1%, *PER_URB_NEIGH*: 34.9%, and *HA_MAX*: 12.5%). Thus, apart from urban areas, high HSI values were associated with high *HA_MAX* and *ROADS* values. However, when correcting the sampling bias effect, the *Landscape (LS)* variable was the second most contributive variable (14.1%), the *ROADS* variable contribution increased to 62.6%, and the *PER_URB_NEIGH* variable contribution decreased to 11.1% (see Table 3).

From a quantitative point of view, the two approaches (with and without applying the correction of the sampling bias effect) resulted in identical AUC and partial AUC ratios. However, the regularized gain and the CBI were lower without correction, with values equal to 2.81 (vs. 3.18) and 0.284 (vs. 0.421), respectively.

Thus, correction of the sampling bias effect gave better results: both more consistent with knowledge from the field and more accurate in terms of prediction. The fact that the contributions of the *LS* and *HA_MAX* variables respectively increased and decreased with the use of the biased background, tends to show that the correction method actually manages to counterbalance the over-

representation of inhabited areas (cities, villages, and gold mining areas) in the sampled data. However, the very high contribution of the variable *ROADS* may be a residual effect of sampling bias, as sampling is essentially performed in the vicinity of accessible roads and tracks. Further studies are necessary to objectively and quantitatively assess the actual performance of the proposed methodology for correcting the effect of sampling bias.

Habitat Suitability and Malaria in French Guiana

Alimi et al. (2015) highlighted the utility of SDMs for gaining a better understanding of the geographical range and distribution of vectors for eliminating malaria and preventing outbreaks. The coastal strip in French Guiana is generally malaria free, although some cases resulting from local transmission are regularly diagnosed (Ardillon et al. 2015). This study, as well as that of Vezenegho et al. (2015), shows that the savanna in French Guiana may be highly suitable for *An. darlingi*. In the forest, Pommier de Santi et al. (2016c) found a link between mining, malaria cases, and the presence of *An. darlingi*. Indeed, >74% of malaria cases in French army soldiers were associated with operations to counteract illegal gold mining (Pommier de Santi et al. 2016a). According to the results of the present study, some areas associated with intense gold mining activity, known to be malaria transmission foci, are not necessarily associated with very high HSI values. In the village of Camopi, the annual malaria prevalence was 70% for children younger than seven years of age between 2000 and 2002 (Carme et al. 2005), reaching 100% in 2006 (Hustache et al. 2007). However, only some pixels on the border of the Camopi and Oyapock rivers have high values on the HSI map (area G in Figure 4). This is consistent with the study of Girod et al. (2011), which showed that the number of *An. darlingi* caught in this village was very low relative to the incidence of malaria cases. These findings collectively highlight two important points. First, the HSI map shown in Figure 4 does not correspond to a map of malaria transmission risk. Transmission risk depends on many factors that were not taken into account here, such as the parasitic charge and immunological status of the local population, compositional and

configurational features of the landscape (Stefani et al, 2013; Li et al, 2016), and behavioral factors. Second, this highlights that malaria transmission can occur in areas where there is a very low density of *An. darlingi*. This may be due to the presence of other *Anopheles* species such as *An. (Nys.) nuneztovari* Galbaldón, *An. (Nys.) oswaldoi* Peryassú, *An. (Nys.) intermedius* Peryassú, *An. (Nys.) marajoara* Galvão and Damasceno, or *An. (Nys.) ininii* Sènevet and Abonnenc (Diptera: Culicidae), already known to be naturally infected with *Plasmodium* species and/or described as efficient secondary malaria vectors (Dusfour et al. 2012, Pommier de Santi et al, 2016c).

Environmental Characterization

A significant limitation of this study was the spatial resolution of the environmental data. Capture campaigns are generally carried out at a local scale (villages or camps; Vezenegho et al. 2015, Dusfour et al. 2013). The spatial resolution of the study was not sufficient to take into account the heterogeneity of the environment at the capture scale. The use of environmental data with higher spatial resolution, such as the canopy height estimation from Fayad et al. (2014) or finer characterization of the land cover could improve future studies. However, these data are not consistently available across the entire territory.

In conclusion, the results of this study help to complete our knowledge on the spatial distribution of the principal malaria vector in this Amazonian region, and to identify the main factors that favor its presence. These results can be exploited to define the necessary targeted vector control strategies in a malaria pre-elimination context, and to extrapolate the acquired knowledge to other Amazonian contexts. They also suggest areas that need to be targeted to complete the field knowledge, validate the prediction and strengthen the model. Eventually, these proposed methodological developments can be applied to other species, including other disease vectors.

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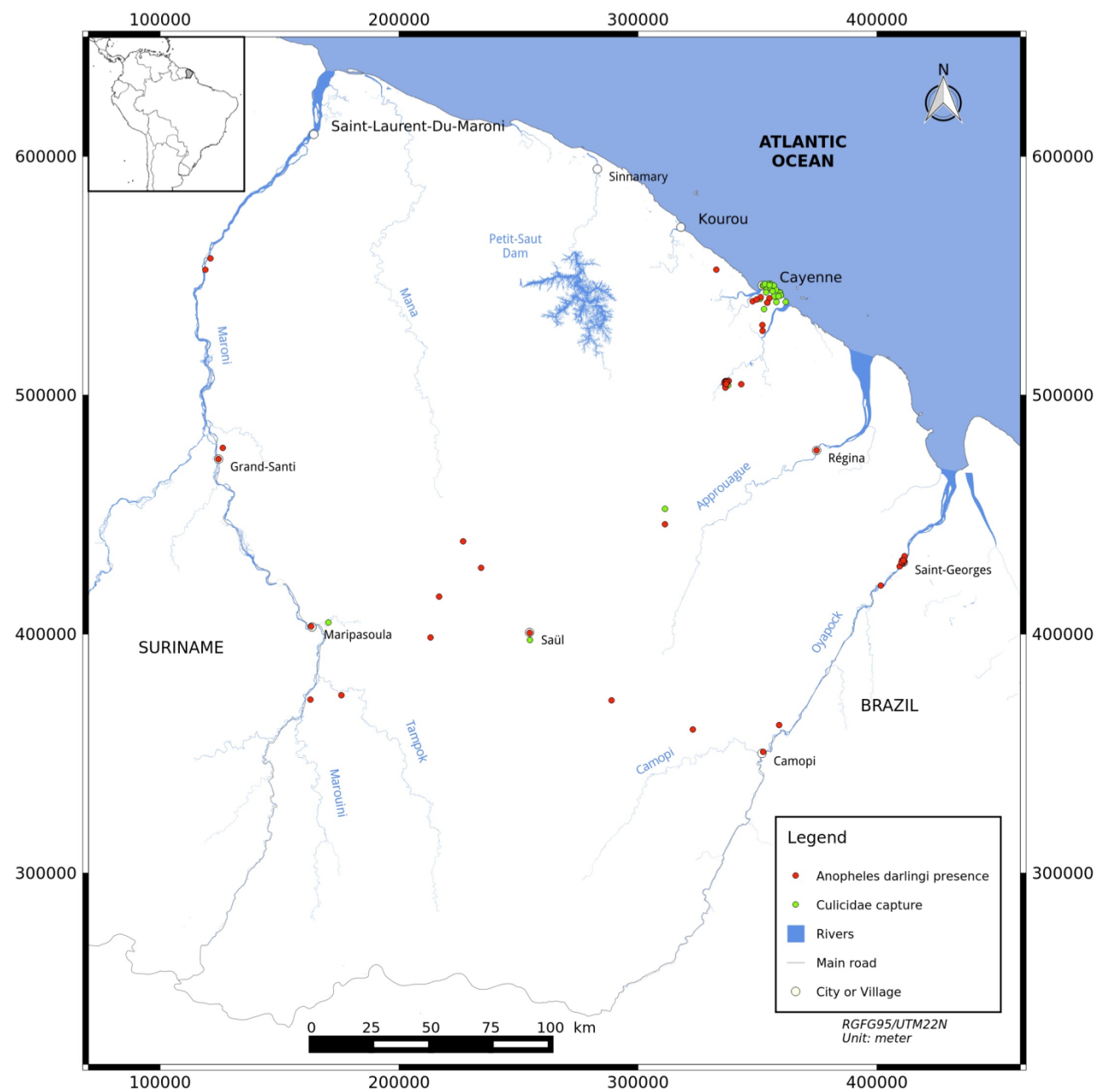


Figure 1. *Culicidae* capture points and *Anopheles darlingi* presence points (from 2000 to 2013).

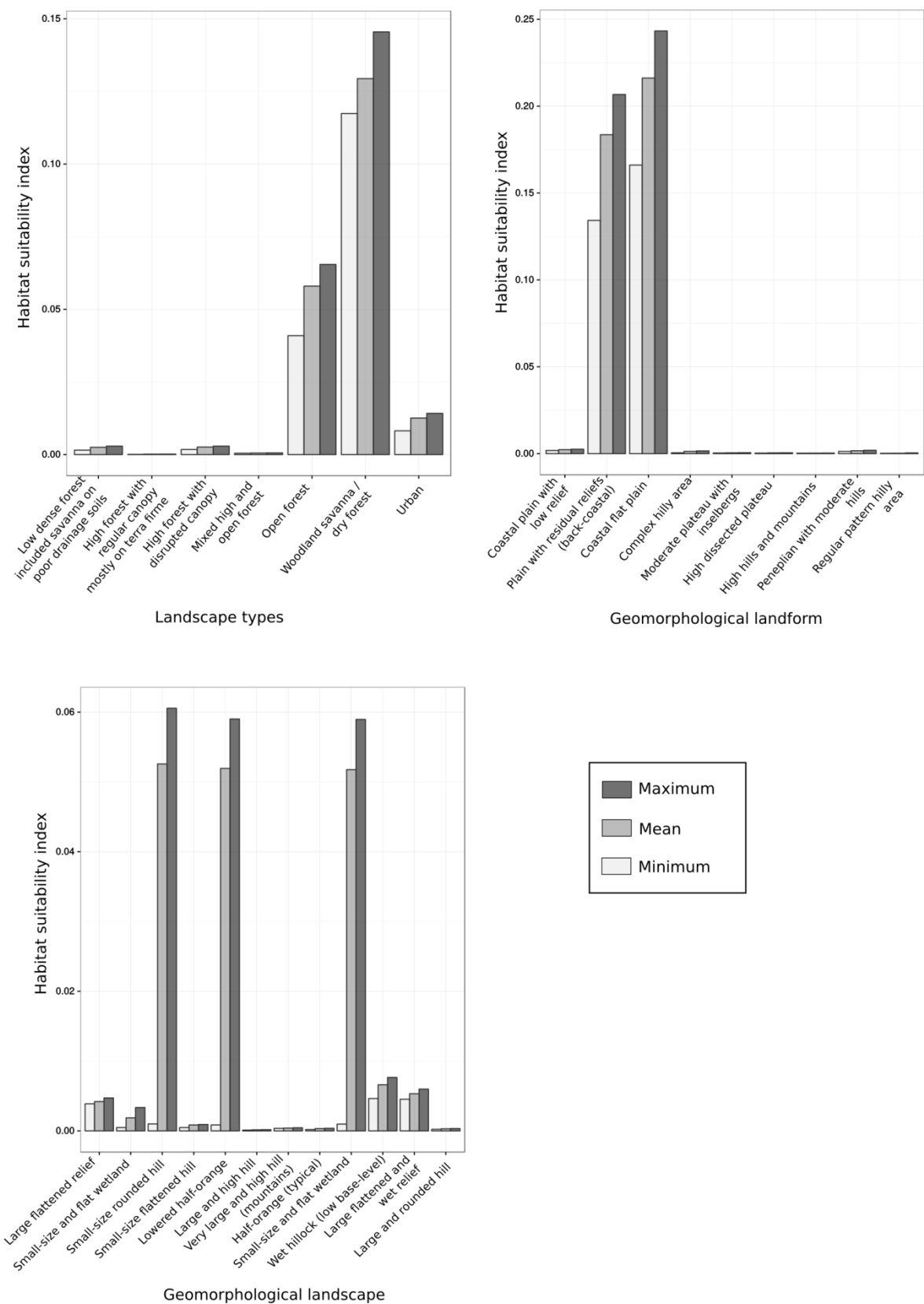


Figure 2. Response curves of categorical environmental variables.

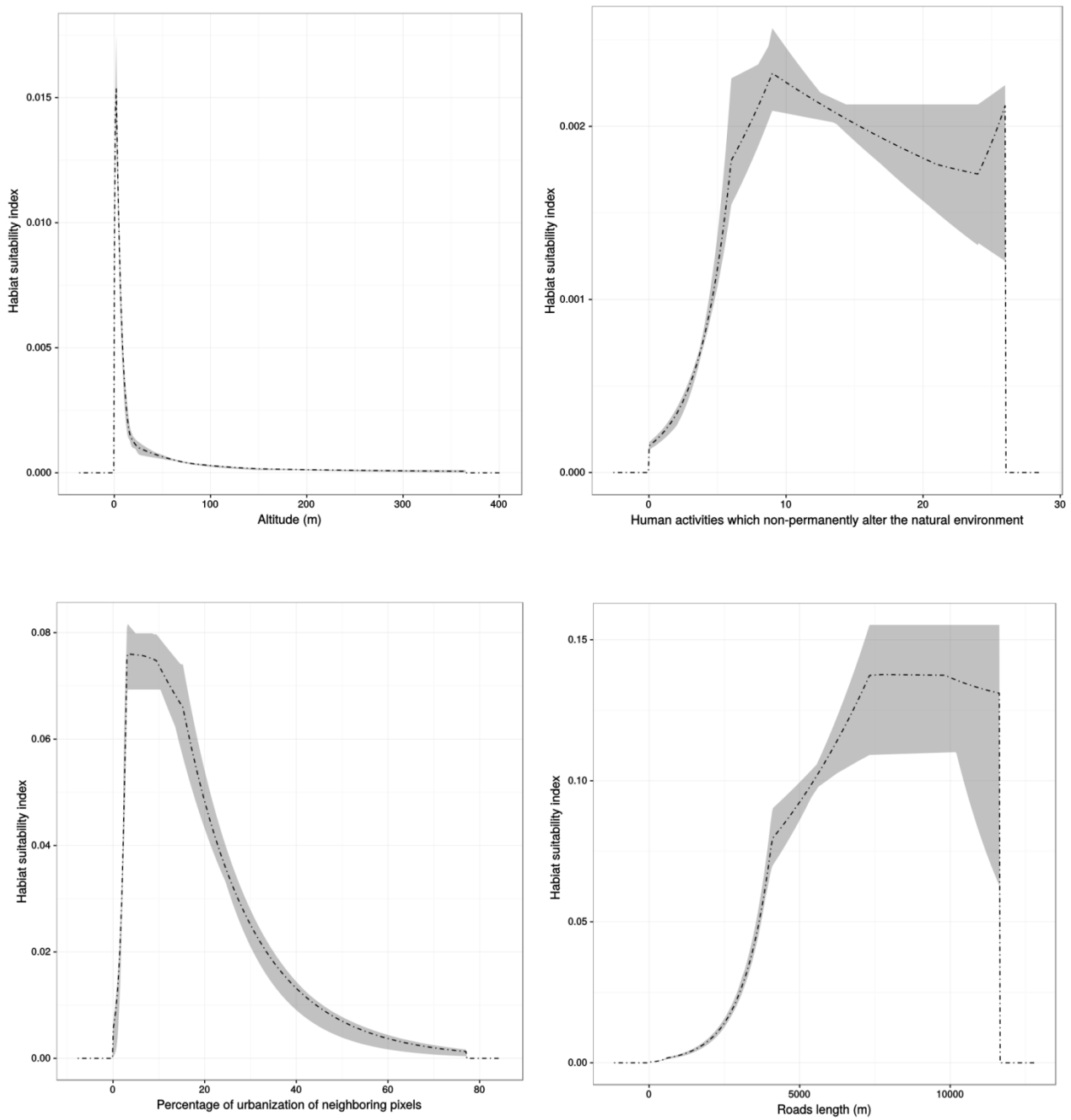


Figure 3. Response curves of numerical environmental variables. Dashed lines show the mean values and the grey regions represent the interval between the maximum and minimum values.

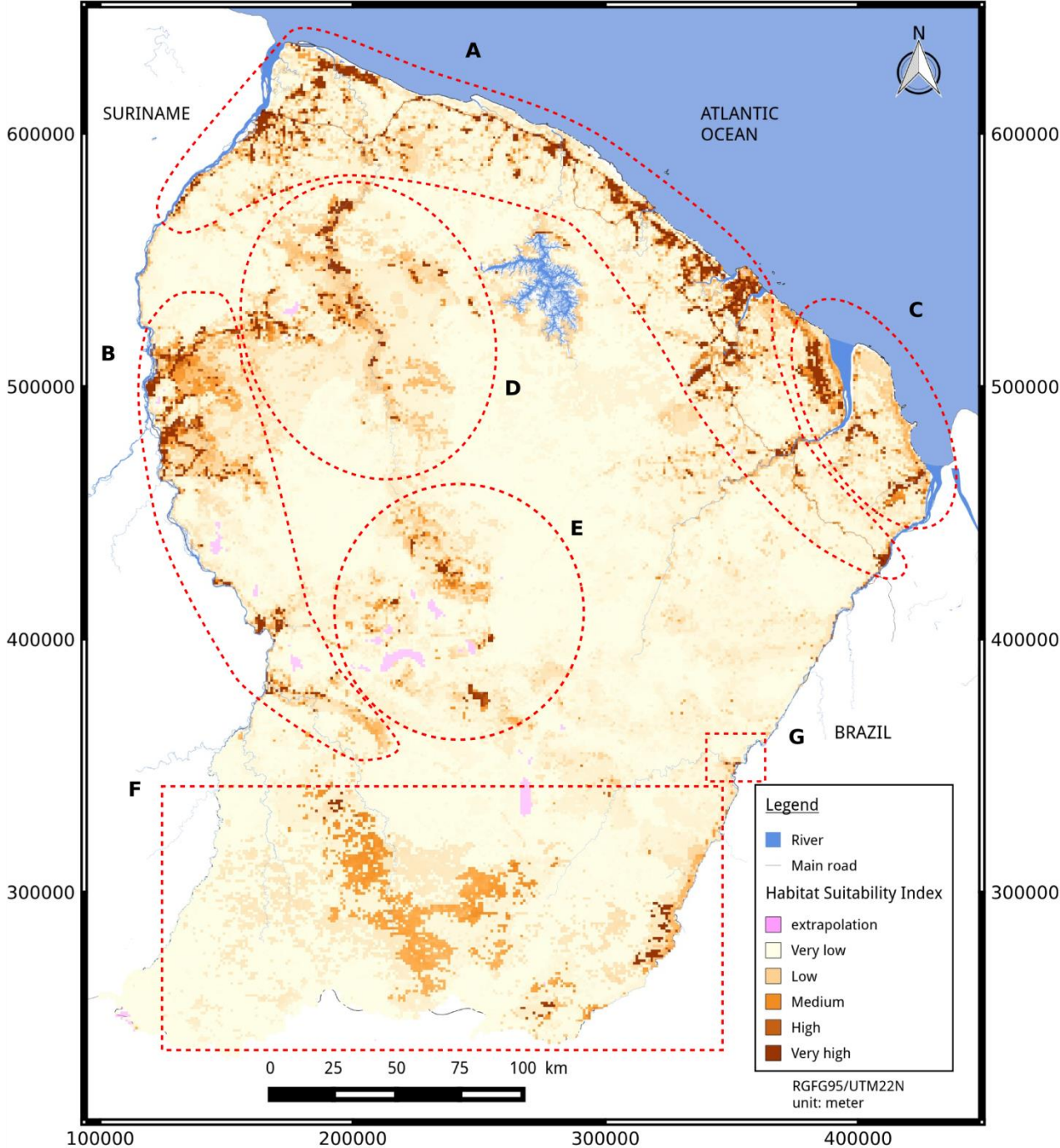


Figure 4. Habitat suitability index map. Six main areas with a high habitat suitability index (A to F) and Camopi village (G) are circumscribed by the red circles and rectangles.

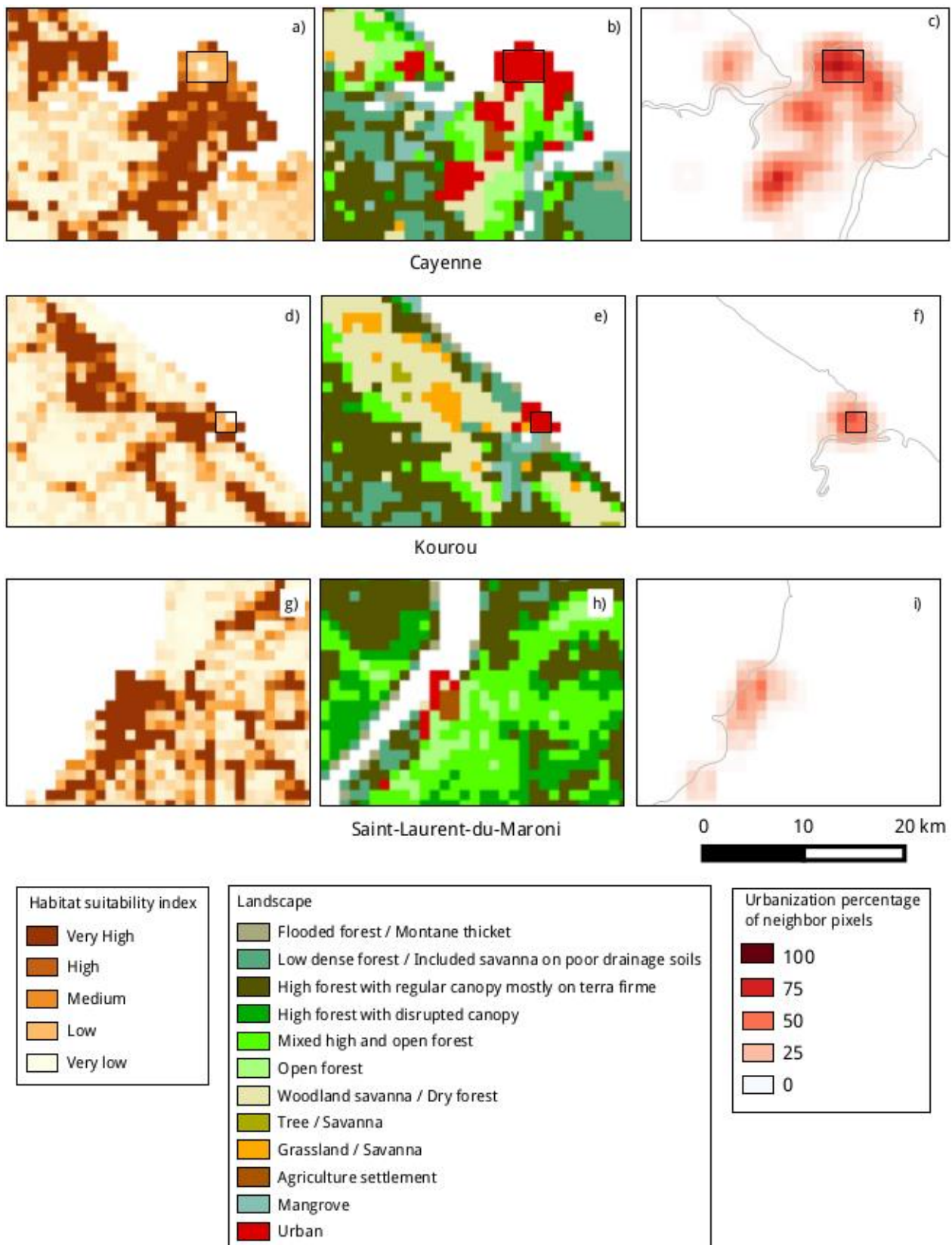


Figure 5. Zoom of urban areas. a, d, and g: habitat suitability index maps. b, e, and h: landscape type. c, f, and i: percentage urbanization of neighbor pixels. Rectangles correspond to highly urbanized areas (*LS* class is *Urban* and *PER_URB_NEIGH* $\geq 50\%$). Cayenne and Kourou include highly urbanized areas, but Saint-Laurent-du-Maroni does not.

Table 1. Raw environmental data and derived variables used to build the model.

Number of input variable	Producer, reference	Raw environmental data	Derived from (information source)	Date(s)	Original spatial resolution or interpretation scale	Derived SDM input variable(s)	Type of feature extraction for each 1x1 km pixel	Classes or range of values and units	Environment types ²	<i>A priori</i> effect on <i>An. darlingi</i> presence ³ and bibliographic references	Input variable(s) type
1	Forest National Office (ONF), (Guitet et al. 2013)	Geomorphological landscape (<i>GLS</i>)	SRTM	2000	≥ 5000 m	Geomorphological landscape (<i>GLS</i>)	Majority class	12 classes	Natural environment	(/) Smith et al. (2013)	Categorical
2	Forest National Office (ONF), (Guitet et al. 2013)	Geomorphological landform (<i>GLF</i>)	SRTM	2000	≥ 200 m	Geomorphological landform (<i>GLF</i>)	Majority class	15 classes	Natural environment	(/) Smith et al. (2013)	Categorical
3	Agricultural Research Centre for International Development (CIRAD), (Gond et al. 2011)	Landscape types (<i>LS</i>)	Spot-Vegetation	2000	1000 m	Landscape types (<i>LS</i>)	Correction of pixels corresponding to urban areas and mangroves	14 classes	Natural environment and urbanization	(/) Stefani et al. (2013) Girod et al. (2011) Zeilhofer et al. (2007) Rozendaal (1992) Hiwat et al. (2010) Vittor et al. (2006) Vittor et al. (2009)	Categorical
4, 5, 6	National Aeronautics and Space Administration (NASA)	Altitude (<i>ALT</i>)	SRTM	2000	30 m	Altitude - minimum (<i>ALT_MIN</i>), - maximum (<i>ALT_MAX</i>) - median (<i>ALT_MED</i>)	Statistical computation	0 – 832 m	Natural environment	(-) Zeilhofer et al. (2007)	Continuous

Number of input variable	Producer, reference	Raw environmental data	Derived from (information source)	Date(s)	Original spatial resolution or interpretation scale	Derived SDM input variable(s)	Type of feature extraction for each 1x1 km pixel	Classes or range of values and units	Environment types ²	<i>A priori</i> effect on <i>An. darlingi</i> presence ³ and bibliographic references	Input variable(s) type
7	National Institute of Geographic and Forestry Information (IGN)	Road and track network	BD TOPO®	2011	≥ 1000 m	Length of roads and tracks outside of urban areas (<i>ROADS</i>)	Computation of road/track lengths	0 – 12545 m	Non-permanent anthropogenic changes	(+) Singer and Castro (2001)	Continuous
8	Association Kwata 'Study and Conservation of French Guianan Wildlife ' (de Thoisy et al. 2010)	Human footprint (HFP)	From various sources ^a	2005	≥ 1000 m	Percentage of urbanization of neighboring pixels (<i>PER_URB_NEIGH</i>)	Percentage of urbanization within the eight neighbor cells	0-100%	Urbanization	(-) Stefani et al. (2013)	Continuous
9, 10, 11	Association Kwata 'Study and Conservation of French Guianan Wildlife ' (de Thoisy et al. 2010)	Human footprint (HFP)	From various sources ^a	2005	≥ 1000m	Human activities which non-permanently alter natural environment (<i>HA</i>) - minimum (<i>HA_MIN</i>) - maximum (<i>HA_MAX</i>) - median (<i>HA_MED</i>)	Statistical computation	0-30	Non-permanent anthropogenic changes	(+) Vittor et al. (2009)	Continuous

^a French Institute for Statistical and Economic studies (INSEE); Regional Departments for Food, Agriculture and the Forest (DAAF); ONF; Regional Equipment, Habitat and Planning Authority (DDE) and Hammond et al. (2007).

^b See the section on environmental variables in Materials and Methods.

^c *A priori* effect on *An. darlingi* presence: (+) favorable; (-) unfavorable; (/) depends on categorical variable values.

Table 2. Mean contributions and jackknife results of the eleven input environmental variables.

Environmental variables	Contribution (%)	Cumulative contribution (%)	Gain with the variable only	Decrease of the gain without the variable (%)
<i>ROADS</i>	51.45	51.45	2.20	-7.98
<i>PER_URB_NEIGH</i>	17.17	68.62	1.86	-0.41
<i>LS</i>	15.32	83.94	2.23	-4.67
<i>HA</i>	7.43 (min: 0.35; median: 0.24; max: 6.84)	91.37	min: 0.02 median: 0.15 max: 0.43	min:-0.06 median: -0.22 max: -2.10
<i>GLS</i>	5.35	96.72	1.40	-2.59
<i>ALT</i>	2.09 (min: 1.34; median: 0.69; max: 0.06)	98.81	min: 1.12 median: 1.04 max: 0.76	min: - 1.04 median: -0.39 max: -0.03
<i>GLF</i>	1.19	100	0.80	-0.21

Table 3. Mean contributions and jackknife results of the seven input environmental variables of the simpler model.

Environmental variables	Contribution (%)	Cumulative contribution (%)	Gain with the variable only	Decrease of the gain without the variable (%)
<i>ROADS</i>	62.61	62.61	2.31	-8.61
<i>LS</i>	14.10	76.71	2.35	-6.23
<i>PER_URB_NEIGH</i>	11.15	87.86	2.05	-0.58
<i>HA_MAX</i>	5.39	93.25	0.37	-1.74
<i>GLS</i>	3.84	97.09	1.44	-1.90
<i>GLF</i>	2.1	99.19	1.01	-0.32
<i>ALT_MIN</i>	0.88	100	1.27	-1.29

Table 4. Characterization of areas with a high HSI

ns. signifies that the high HSI of the concerned area was not driven by that environmental variable, (+) signifies that when the value of the variable increases, the HSI also increases also, (-) signifies that when the value of the variable decreases, the HSI increases, and cells with classes name signifies that the presence of the given class implies a high HSI.

Area	ROADS	LS classes	PER_URB_NEIGH	HA_MAX	GLS classes	GLF classes	ALT
A	(+)	- Woodland savanna / Dry forest - Mixed high and open forest	(-)	(+)	- Coastal plain with low relief - Plain with residual reliefs (back coastal)	- Small size and flat wetland - Large flattened and wet relief - Wet hillock (low base-level)	(-)
B	(+)	- Open forest - Mixed high and open forest	<i>ns.</i>	(+)	- Peneplain with moderate hills	-Wet hillock (low base-level) - Large flattened relief	(-)
C	<i>ns.</i>	- Open forest	<i>ns.</i>	<i>ns.</i>	- Coastal flat plain	- Large flattened and wet relief	(-)
D	(+)	- Mixed high and open forest	<i>ns.</i>	(+)	<i>ns.</i>	<i>ns.</i>	(-)
E	(+)	- Mixed high and open forest	<i>ns.</i>	(+)	- Peneplain with moderate hills	- Large flattened relief	(-)
F	<i>ns.</i>	- Open forest	<i>ns.</i>	<i>ns.</i>	- Peneplain with moderate hills	- Lowered half-orange	(-)
G	(+)	Mixed high and open forest	<i>ns.</i>	(+)	<i>ns.</i>	<i>ns.</i>	(-)

S1. Coordinates of *Anopheles darlingi* presence sites (Coordinate system: RGFG95/UTM22N)

Number of sites	Locality	Longitude	Latitude
1	Cayodé	175866.785122426	374453.1653833
2	Taluène	162884.684665956	372648.888139721
3	Cacao	336895.086300153	505521.800163598
4	Midenangalanti	121041.015986301	557342.72581239
5	Grand Santi	124374.105534813	473261.338110987
6	Bois Martin	118927.138507594	552587.496776593
7	Flavien Campou	126212.611562883	477960.065899793
8	Régina	374698.048396995	476928.730177523
9	Camopi	352395.046055802	350764.216480451
10	Alikéné	322929.978284572	360135.527835386
11	Mine Boulanger	343189.582357019	504590.723307373
12	Carbet Légion crique Sikini	359046.553479	361996.097871
13	Cogneau - 23. Lot. Aquavilla	354091.053713437	538837.801956188
14	41. rue des Ixoras - Lot. Cogneau Larivot	351322.960529913	541116.10235073
15	Attila Cabassou	354987.967015231	540678.87835953
16	La Chaumière	347961.540051952	539402.864452834
17	1228. Ch. de La Chaumière	349626.610158034	540136.658859129
18	Saint-Georges	411052.118458134	429833.606330522
19	Quartier Espérance - Saint-Georges	411113.960068634	430709.665437749
20	Village Martin - Saint-Georges	411519.804866871	432628.481492214
21	Boulangerie – Saül	254644.262333224	400558.641871421
22	Chemin Mogès	352032.329517247	529456.717653458
23	Dorlin	216742.863044	415732.862195
24	Maripasoula	163199.637191727	403271.422431979
25	Repentir	234327.380239	427769.677349

Number of sites	Locality	Longitude	Latitude
26	Stoupan	352127.057631364	527017.037636427
27	Camp Pararé – Nouragues	311267.546406115	446010.858172732
28	Village Blondin - Saint-Georges	409492.529993713	428392.815435875
29	Quartier Adimo - Saint-Georges	410525.149504269	430989.662548913
30	Camp Bernet/ Légion étrangère	410437.343147834	429860.701524782
31	La ferme de Lait-Quateur	332757.212761399	552628.404024885
32	Grand Usine	288898.531939378	372330.073251345
33	Dagobert	226797.35968	438864.200286
34	Cacao	336515.199831583	505801.969900718
35	Saut-Maripa – Camp militaire	401579.145107468	420354.782186144
36	Eau-Claire	213145.824694	398626.838699
37	Cacao	336956.199791048	504651.969979654
38	Cacao	337909.199733996	506008.969874801
39	Cacao	336576.199808207	503413.970071365
40	Cacao	336555.199807874	503184.970087929
41	Impasse de la raffinerie – Cogneau	354091.053713437	538837.801956188
42	Quartier Bambou	411277.041477492	430178.275876772
43	Quartier Maripa - Saint-Georges	410252.974346512	430188.463058511
44	Quartier Savane - Saint-Georges	411024.737954334	430940.054473617
45	Cacao	336826.199809151	505764.969900894
46	Cacao	337091.199787723	505441.969921949
47	Cacao	336186.199848992	505049.969957266
48	Quartier Onozo- Saint-Georges	411339.040785181	430641.896115415

S2. Creation of the relative sampling effort map

Capture data of *Culicidae* (74 capture sites) were used to estimate the sampling effort of *An. darlingi*. The collection methods were identical and the sampling bias for the family was assumed to be representative of that for the focal species.

The sampling bias was defined as the relative sampling effort in the environmental space. For a pixel i , it corresponds to the ratio of the number of sampled pixels over the total number of pixels, within the *environmental neighborhood* of i .

First, all pixels of the study area were represented in the environmental variable space. This was accomplished by performing a Factorial Analysis of Mixed Data (FAMD) (Pagès, 2004). This analysis jointly takes into account numerical and categorical variables and makes it possible to represent the pixels within an Euclidean, orthonormal space defined from the whole set of environmental variables.

The membership degree of a pixel j to the neighborhood of pixel i , denoted w_{ij} , was defined by a Gaussian-like membership function:

$$w_{ij} = 0.5 \left(d_{ij} / D_{min} \right)^2 \quad (1)$$

with d_{ij} the euclidean distance between i and j in the factorial space, and D_{min} the threshold distance over which j does not significantly belong to the environmental neighborhood of i , *i.e.* over which $w_{ij} < 0.5$. The membership degree w_{ij} has the following properties:

- $w_{ij} \in]0,1]$;
- $w_{ij} = 1$ if $d_{ij} = 0$;
- $w_{ij} < 0.5$ if $d_{ij} > D_{min}$.

The parameter D_{min} was set from *a priori* knowledge of *An. darlingi* bio-ecology. As highly urbanized areas are not suitable for *An. darlingi* (see § I.3), we stated that a pixel associated with *An. darlingi* presence cannot belong to a highly urbanized pixel. Reciprocally, a pixel considered to be highly urbanized cannot belong to the environmental neighborhood of a pixel where *An. darlingi* was observed.

Consequently, given P , the set of pixels where the species was observed and U , the set of pixels belonging to highly urbanized areas, D_{min} was defined as follows:

$$D_{min} = \min(d_{pu})_{p \in P, u \in U} \quad (2)$$

A pixel is considered to be highly urbanized if it belongs to the *LC* class *Urban* and if its eight neighboring pixels present an average urbanization percentage (*PER_URB_NEIGH*) higher than or equal to 50%.

The concepts of environmental space and neighborhood, as well as the key method parameters are schematically represented in Figure S1.

Given X , the set of pixels of the study area, and $c = \{c_i\}_{i \in X}$, a vector such that $c_i = 1$ if i is sampled and $c_i = 0$ otherwise, the relative sampling effort at pixel i , z_i , is then defined as:

$$z_i = \sum_{j \in X} w_{ij} \cdot c_j / \sum_{j \in X} w_{ij} \quad (3)$$

The relative sampling effort was computed for each pixel of the study area. The resulting map was used to bias the random selection of background points. Consequently, for a given pixel, the greater the relative sampling effort, the higher the chance of selecting the pixel as a background point.

Reference

Pagès, J. 2014. Multiple Factor Analysis by Example Using R. Chapman & Hall, CRC Press.

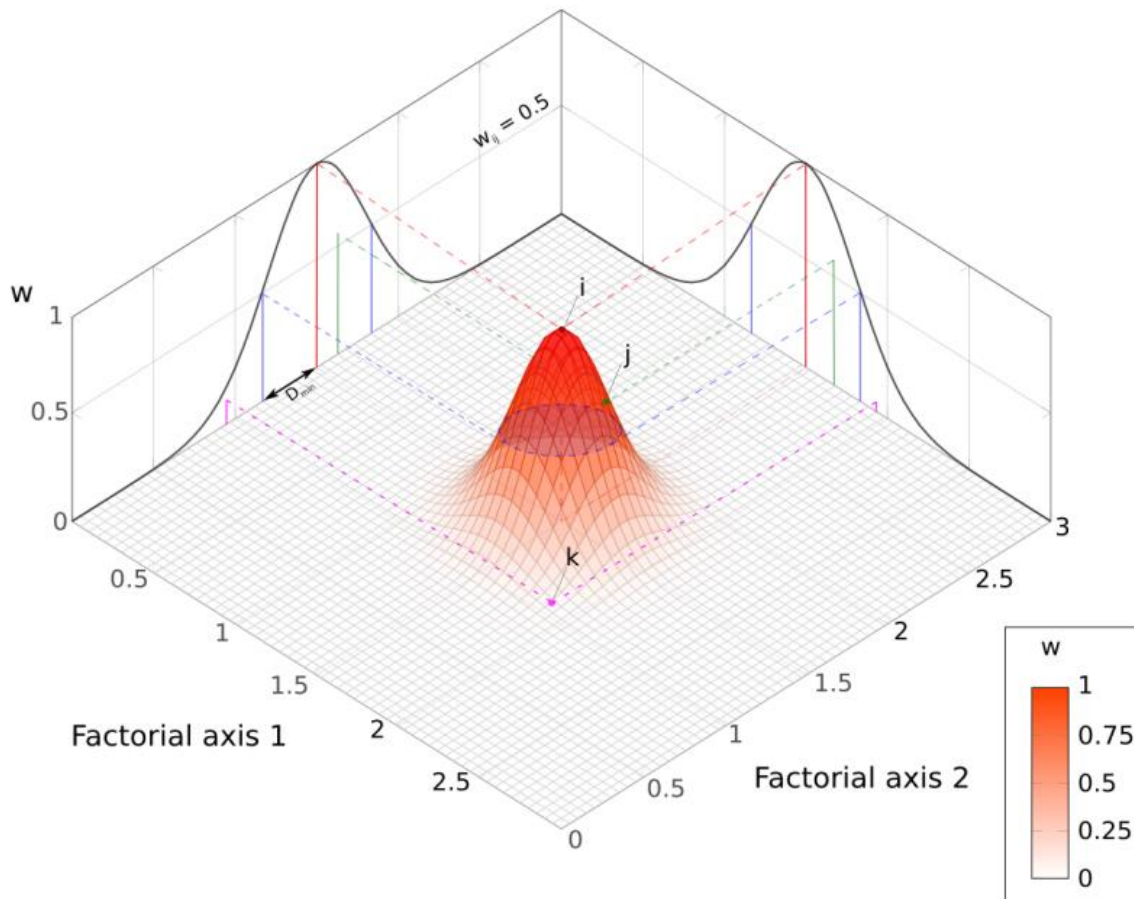


Figure S1. Neighborhood of a pixel i in the environmental space represented by the first and second factorial axes. The environmental neighborhood of point i is represented by the Gaussian function. The blue lines define the limit of the neighborhood of i . Only point j is situated above these lines. Thus j is in the neighborhood of i in the first factorial plane.